

DTSA 5751

# An Introduction to Molecular Simulation Methods and Tools

|                                                    |                                |               |
|----------------------------------------------------|--------------------------------|---------------|
| FORMAT                                             | CATEGORY                       | CREDITS       |
| Online (DTSA)                                      | Elective                       | 1             |
| SECTION                                            | BEST ROLE MATCH                | AVG ALIGNMENT |
| Section 4 – Elective                               | Applied Scientist (63.9 / 100) | 47.3 / 100    |
| SPECIALIZATION Simulation Methods for Data Science |                                |               |

47

CURRENT AVG SCORE  
across 6 job roles

84

POTENTIAL SCORE  
with recommended additions

+45

MAX SCORE GAIN  
best single-role improvement

CURRENT COURSE DESCRIPTION

This course will cover the theory and practical approaches to modeling simple molecular systems described by classical physics. It will cover all the fundamentals needed to go from a description of a system at a molecular level through the generation of valid simulation data all the way through analysis of the simulation data to produce useful results. The course will cover the most common open-source simulation programs as well as Python-based analysis tools.

INDUSTRY ALIGNMENT BY JOB ROLE

| JOB ROLE                  | CURRENT SCORE               | PROJECTED SCORE              | POINT GAIN | PRIORITY |
|---------------------------|-----------------------------|------------------------------|------------|----------|
| Applied Scientist         | <div><div></div></div> 63.9 | <div><div></div></div> 100.0 | +36.1      | HIGH     |
| AI Engineer               | <div><div></div></div> 51.6 | <div><div></div></div> 89.2  | +37.6      | HIGH     |
| Machine Learning Engineer | <div><div></div></div> 49.9 | <div><div></div></div> 87.5  | +37.5      | HIGH     |
| Data Scientist            | <div><div></div></div> 46.4 | <div><div></div></div> 67.6  | +21.2      | MEDIUM   |
| ML Engineer               | <div><div></div></div> 40.6 | <div><div></div></div> 84.2  | +43.7      | HIGH     |
| MLOps Engineer            | <div><div></div></div> 31.2 | <div><div></div></div> 75.7  | +44.6      | HIGH     |

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold ( **example** ) appear as gaps across 3 or more job roles.

**LLMs & Generative AI** 1 of 6 roles

storage

**Business Communication** 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

LOW

**Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.
- 2

LOW

**Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

**Applied Scientist**

63.9 → 100.0 (+36.1 pts)

computational storage problems models technical drug learning design

**AI Engineer**

51.6 → 89.2 (+37.6 pts)

models learning infrastructure build software technical model based

**Machine Learning Engineer**

49.9 → 87.5 (+37.5 pts)

learning solutions exploratory business design development science models

**Data Scientist**

46.4 → 67.6 (+21.2 pts)

social build problems

**ML Engineer**

40.6 → 84.2 (+43.7 pts)

media learning models model infrastructure build problems technical

**MLOps Engineer**

31.2 → 75.7 (+44.6 pts)

infrastructure model text learning media models development

**Methodology:** Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.